



**Verified Carbon
Standard**

DELAGUA IMPROVED COOKSTOVE GROUPED PROJECT



Document Prepared by DelAgua

Project Title	DelAgua Improved Cookstove Grouped Project
Version	1.0
Date of Issue	05-September-2022
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1 PROJECT DETAILS

1.1 Summary Description of the Project

DelAgua clean cooking grouped project 3699 is a voluntary Grouped Project (GP) aimed at distributing fuel efficient improved cookstoves (ICS) to households in Rwanda. The Project Participant (PP) of the GP for all the communication, management, and ownership rights of the Greenhouse Gas (GHG) Emission Reduction Units¹ (ERU) is DelAgua Health Rwanda (Voluntary) Limited (DelAgua) and the ICS under this GP will be distributed by DelAgua. The distribution of ICS is not mandated by Rwandan law and the Grouped Project is a voluntary initiative being run by the PP DelAgua. The program is not being implemented as a result of any government legislation or intervention. Throughout this joint Project Description & Monitoring Report (PD & MR), this grouped project may be interchangeably referred as “program”.

The ICS design has a stainless-steel insulated combustion chamber, a small insulated sub-chimney and air inlets, which allows constant air inflow and improves combustion efficiency. Thermal transfer from the stove to the cooking pot is increased and GHG emissions are reduced.

Under the GP, DelAgua will distribute the project technologies to individual households and communities identified in several regions within the host country Rwanda. The PAIs will occur in geographically defined regions within Rwanda. The details of the GP and PAI locations are provided in Section 1.12 of this document.

To calculate the GHG ERUs (VCUs) arising out of the ICS, this GP will follow the VCS Methodology from Sectoral Scope 3 - VMR0006 “Methodology for Installation of High Efficiency Firewood Cookstoves”, version 1.1. The fuelwood savings arising out of the usage of project ICS will be counted towards the amount of GHG emissions reduced when compared with the project’s baseline.

The scenario existing prior to the implementation of the project activity would be usage of 3 stone fire and traditional stoves, which have a very low thermal efficiency, and therefore consume a higher amount of firewood as fuel for cooking.

The average annual and total GHG emission reduction expected from the grouped project is expected to be 328,201 and 2,297,407 tCO₂e, respectively, over the first 7-year crediting period.

1.2 Sectoral Scope and Project Type

¹ Tradeable Market based Instrument under the Paris Agreement, where ERUs are following the definition set under the UNFCCC Kyoto Protocol, “1 ton equivalent of CO₂ removed or avoided”, more appropriately known as the CER under CDM, & Voluntary Carbon Unit under VCS.

Sectoral Scope 03 – Energy demand

Type II – Energy efficiency improvement projects

Activity category – non AFOLU

The project activity is designed as a Grouped Project Activity (GP) and includes several Project Activity Instances (PAIs) of the same activity type into one Grouped Activity. New instances can be introduced to the GP at any monitoring period, with a PAI defined as:

- Distribution of efficient cookstoves to a local community within a specific geographic and climatic region in the project boundary

1.3 Project Eligibility

The Grouped Project Activity falls within the scope of the VCS Program. Also, none of the two activities within the GP are listed in the category of excluded projects in Table 1 of the VCS Standard ver. 4.2, and therefore, it is eligible under the scope of the VCS Program.

Table 1: VCS Program scope

Scope	Applicability
The six Kyoto Protocol greenhouse gases.	The GP reduces GHG emissions associated with the combustion of cooking fuels. Gases included are CO ₂ , CH ₄ , N ₂ O.
Ozone-depleting substances (ODS).	The GP does not involve any ODS.
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The ICS distributed (& to be distributed) in the GP uses the methodology “VMR0006 <i>Methodology for Installation of High Efficiency Firewood Cookstoves</i> ”, version 1.1.” which is a VCS approved methodology.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	Not applicable.

Table 2: Excluded project activities under the VCS Program

Excluded Activity	Applicability
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Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A. The GP reduces GHG emissions associated with the combustion of cooking fuels
Grid-connected electricity generation using hydro-power plants/units	N/A The GP activities are energy efficiency measures and do not generate electricity from hydro-power plants/units.
Grid-connected electricity generation using wind, geothermal, or solar power plants/units	N/A. The GP activities are energy efficiency measures and do not generate electricity from wind, geothermal, or solar power plants/units.
Utilization of recovered waste heat for, <i>inter alia</i> combined cycle electricity generation and the provision of heat for residential, commercial, or industrial use	N/A. The GP activities are energy efficiency measures and do not utilize recovered waste heat for any purposes.
Generation of electricity and/or thermal energy using biomass. This does not include efficiency improvements in thermal applications (e.g., cookstoves)	N/A. The GP activities are energy efficiency measures and do not generate electricity using biomass. The GP includes energy efficiency improvements in thermal applications, i.e. cookstoves.
Generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel	N/A. The GP activities do not include the generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel.
Replacement of electric lighting with more energy efficient electric lighting such as the replacement of incandescent electrical bulbs with CFLs or LEDs	N/A. The GP activities do not include the replacement of electric lighting.

Installation and/or replacement of electricity transmission lines and/or energy efficient transformers	N/A. The GP activities do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.
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1.4 Project Design

When completing a draft project description for the purpose of listing on the pipeline as under development, complete the following information; otherwise, delete this text:

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ The project is a grouped project

Eligibility Criteria

Table 3: Eligibility Criteria for Distribution of ICS within the project boundary

S No	Criterion	How the new project activity instances comply	Evidence
1	Project Activity Instances (PAIs) must meet the applicability conditions set out in the applied methodology.	<i>Please refer the subsections below.</i>	<i>Please refer the subsections below.</i>
1a	Project activities shall be implemented in domestic premises or in community-based kitchens.	Each PAI will distribute technology that is appropriate for domestic use to household end-users	a) Manufacturer's specifications. b) Project ICS distribution database containing unique household end user information.

1b	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	ICS distributed in each PAI will have a thermal efficiency of at least 25%, and will be confirmed by the end user that they were using the firewood as the cooking fuel in the project's baseline. Each PAI will record the baseline and project fuel use to ensure that wood fuel is the sole source of energy.	1. Manufacturer's specifications. 2. Information recorded within the baseline KPT and project monitoring surveys on the fuel usage.
1c	Both 'Projects' and 'Large Projects' can use the methodology	PAIs will comply with the GP requirements	The GP is defined as a "Large Project" according to the guidelines set out in the VMR0006 methodology and can therefore apply the methodology.
1d	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Each PAI will demonstrate that Non-renewable biomass has been used in the project region since 31 December 1989.	Publicly available studies ^{2,3} are being used to ascertain biomass usage for the PAI within the GP.
1e	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable	Biomass residues are not processed as a fuel to be used in the GP. End users will be asked to state their source of energy and baseline appliance at the point	1. Monitoring surveys. 2. Project ICS distribution poster specific to the end-users (sample photographs)

² <https://irpj.euclid.int/articles/the-transition-from-cooking-with-charcoal-to-lpg-gas-in-rwanda-2018-2024/#:~:text=For%20centuries%2C%20the%20Rwandan%20population,problem%20started%20with%20rapid%20deforestation.>

³ The usage of firewood as fuel for cooking in years prior to 1989 is mentioned at multiple instances within the research thesis, along with the relevant references. <https://edepot.wur.nl/272116>

	biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	of distribution and during the project monitoring. Above information will be stored in the monitoring database.	
1f	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	To avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient Each PAI will have system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including: 1. Date of distribution	1. Project (GP) Distribution database can be checked to verify the distribution approach, alongwith the distribution system (includes both plan and execution steps) established by the PP. The distribution database is an electronic document system. 2. The VCS GP ICS units will have unique engravings or text specifically mentioning the unit is strictly “not for sale” alongwith the DelAgua logo.

		2. A unique serial number⁴ on each technology 3. Available end-user data: name, phone number, address/physical location, ID number	
1g	The Project Design Document shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The End User Agreement confirms that the ownership rights of the ER/VCUs generated by the PAI reside with the PP. The signature of each end user confirming their understanding of this agreement will be collected at the point of distribution. The End User Agreement will be in English and translated into a local language where required. If necessary, it will be read out to the End User. All rights to distribute the delivered stoves and claim the resultant GHG emissions reductions belong to the PP. The GP will not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.	a) End users will be asked to sign the End User Agreement and this information will be a part of the agreement, and then stored in the monitoring database. b) Each PAI will include a statement on the uniqueness of the Project and by listing other similar activities (from other GHG programme registries) that are taking place within the project boundary to demonstrate its uniqueness.
2	Use the technologies or measures specified in the project description.	Only ICS that conform with the GP description are to be distributed in the project. The	Manufacturer's specifications.

⁴ There are no similar programs running in the host country Rwanda where the same ICS models are distributed by another entity under different program(s). If this is found during the project ICS monitoring or verification, it shall be ensured that double counting is avoided by using distinct nomenclature for the unique ID code of the ICS compared with the different programs, and also making sure that it has a distinct color, label, or marking.

		ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	
3	Apply the technologies or measures in the same manner as specified in the project description.	The ICS distributed in the PAIs will adhere to the GP description and replace biomass burned in the baseline scenario.	Manufacturer's specifications.
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new PAIs will be implemented only in regions within the geographic borders of Rwanda subject to the same baseline scenario determined in Section 3.4 in this Project Description. The geographical location of the project ICS can be traced through the information provided in the distribution database, which includes the location data of the ICS distributed.	ICS Distribution database
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new PAIs will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology.	a) An analysis of Rwandan government programmes or policy for cookstoves will be conducted in the PAI design document. See Section 3.5. b) The End User Agreement will confirm that the PAI "installs or distributes stoves at zero cost to the end-

		Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.	user and has no other source of revenue other than the sale of GHG credits.
6	Where a capacity limit applies to a project activity included in the project, no PAI shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: Each device that exceeds one percent of the capacity limit shall be identified. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	The GP does not have any specified capacity limits applicable, since there are no capacity limits defined within the applied methodology, VMR0006, ver. 1.1. Therefore, this eligibility criteria is not applicable.	N/A

Eligibility conditions specific to inclusion of New PAIs			
7	The new PAI must occur within one of the designated geographic areas specified in the project description	The distribution of the ICS is based on a MoU signed with the Government of Rwanda, and therefore, the project activity only distributes to the beneficiaries that are provided in Ubudehe based list of the districts. There is no way this program/GP can distribute ICS in the boundaries of the neighbouring countries of Rwanda. This requires passing of the project ICS through customs and other procedural regulations.	1. Sample list of the beneficiaries that are to be provided the project ICS will be shared. 2. The MoU with the Rwanda government will be shared.
8	The new PAI must comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	The eligibility criteria set must be complied by the PAIs at all points in time throughout the crediting period. If at any point, it is found that the PAI is not complying with the set of eligibility criteria, then it may be excluded from the crediting for the duration of the monitoring period, until it is able to demonstrate the compliance of the eligibility criteria.	The MR will act as the documentary compliance summary. However, for each specific eligibility criteria (as laid down above), the supporting evidence will be provided by the PP.
9	The new PAI must be included in the monitoring report with sufficient technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	The new PAI will include all the technical, financial, geographic and other relevant information related to the compliance of the applicable set of eligibility criteria & also enable sampling by the VVB.	The MR for the specific monitoring period will act as the supporting evidence for the stated condition/requirement.

10	The new PAI must be validated at the time of verification against the applicable set of eligibility criteria.	The new PAI(s) shall be part of the monitoring, and therefore, will be validated against the criteria specified, by an VCS accredited/approved VVB.	The Verification Report based on the VCS template, shall be issued by the approved VVB. This will act as the supporting evidence against this requirement.
11	The new PAI must have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	The PAI related evidence on the project ownership will be in the form of a distribution end user receipt poster. This will be signed by the end user, and a photograph of him/her holding the poster along with the ICS will be saved in the PAI database.	1. A photograph of the PAI distribution first end user holding the poster along with the ICS. 2. A scan of the poster, which will show the first distribution date. 3. The distribution database of the GP will provide the start date in the form of the first sales made by the PAI.
12	The new PAI must have a start date that is the same as or later than the grouped project start date.	The GP is a retroactive activity, and therefore, the new PAI will have a start date later than the GP start date.	This can be verified through the GP distribution database, wherein each new PAI can be verified for its start date.
13	The new PAI must be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification	The PAI will be only eligible for the crediting from its start date through to the end of the project crediting period (only).	1. The start date of the PAI can be checked through the distribution database. 2. The end date of the GP crediting shall be referenced from the VCS Project Registry webpage. 3. The overlap of monitoring periods may be checked through the

	period (as set out in Section 3.4.4 of the VCS Standard, ver. 4.2 ⁵) and new instances are eligible for crediting from the start of the next verification period.		previous monitoring reports (available publicly on VCS project registry webpage for those periods).
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1.5 Project Proponent

Organization name	DelAgua Health Rwanda (Voluntary) Limited
Contact person	Ajay Dixit
Title	Mr.
Address	c/o Cafferata & Co., Poinciana & West Mall Drove, Freeport, Grand Bahama, P.O. Box F-42614, Commonwealth of the Bahamas
Telephone	+250-790002992
Email	ajay.dixit@delagua.org

1.6 Other Entities Involved in the Project

There are no other entities involved in this Grouped Project activity.

Organization name	N/A
Role in the project	-
Contact person	-
Title	-
Address	-
Telephone	-
Email	-

⁵ The monitoring period of the monitoring report shall be a distinct time period that does not overlap with previous monitoring periods. Projects shall not be eligible for crediting of GHG emission reductions generated in previous monitoring periods. In addition, monitoring periods shall be contiguous with no time gaps between monitoring periods.

1.7 Ownership

DeAgua Health Sierra Leone (Voluntary) Limited is the owner of the VCS grouped project activity.

This will be evidenced through the below:

- DeAgua has executed a Memorandum of Understanding with the Government of Rwanda, whereby DeAgua is assigned responsibility for managing and implementing the project and is recognized as the sole direct recipient of all carbon credits derived from it.
- Manufacturer agreements confirm that DeAgua is the sole owner of all technologies purchased under this project. These agreements further confirm the manufacturer's understanding that the carbon credits associated with use of the technologies
- During the distribution of the ICS, the participating household will sign an End User Agreement to confirm that the ownership rights of the carbon assets generated from this project lie with the project proponent.

1.8 Project Start Date

The start date for this Grouped Project is 02-March-2020, which is only including CPAs 18 to 26 from CDM PoA 9626. This is the first date from which improved cookstoves were distributed retroactively under the PAI-1, and began generating GHG emission reductions. As per the VCS requirements⁶, the start date of the VCS project activities must not be more than 2 years before the date of submission of project validation. However, this case is an exception to the VCS requirements, given the fact that the grouped project activity is:

- 1) Previously registered as CPAs 18 to 26 under CDM PoA 9626.
- 2) The last date of crediting for the CDM PoA for the CPAs 18 to 26 (under 5th monitoring period) was 31-December-2020.

The ICS under the GP were distributed in 2020 (PAI – 1). Retroactive crediting for this PAI is being requested under this joint PD & MR document.

The important dates related to the project timeline are provided below.

- i) 02-March-2020 – Distribution of first batch of ICS under PAI-1 took place.

⁶ Refer VCS Standard version 4.3, Section 3.7.1.

- ii) 31-December-2020 – Last date of issuance of credits for ICS units distributed under the CDM PoA CPAs 18 to 26.
- iii) 01-January-2021 – The rights for the GHG emission reduction units generated by the ICS distributed under the CDM PoA 9626 CPAs (18 to 26) are transferred to a different entity DHRV Limited (from this being the effective date), that does not have any communication rights or relations with the CDM PoA.
- iv) 01-January-2021 – The VCS Grouped Project starts crediting retroactively (for PAI-1), with the first date of operation of ICS under DHRV Limited being 01 January 2021.

1.9 Project Crediting Period

The project crediting period will be seven years, twice renewable, for a maximum total of 21 years.

The first crediting period will be 01-January-2021 to 31-December-2027, including both the dates.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	☒

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2021	404,784
2022	391,761
2023	376,153
2024	365,568
2025	349,451
2026	338,122
2027	71,568
Total estimated ERs	2,297,407
Total number of crediting years	7 (renewable twice)

Average annual ERs

328,201

1.11 Description of the Project Activity

Project Activity brief on retroactive distribution:

The grouped project activity, being a retroactive activity, has already implemented PAI-1, for which ICS were distributed in 2020.

Following points will highlight the retroactive distribution:

The ICS distributed under PAI-1 involved Dura model ICS and had a thermal efficiency well above the threshold value of 25%, which is stated as a requirement in Section 4 of the methodology. The efficiency value stayed above this value throughout the retroactive crediting period.

Figure below provides the details of the Dura ICS model that was distributed under PAI-1 and are planned to be verified by the VVB on-site at the time of first Verification (Joint Validation & Verification). Along with each ICS unit, a pot skirt is also provided to the end-users, so that the maximum efficiency of the ICS can be attained at the time of cooking.

Figure 1: Dura ICS, & pot skirt distributed under the GP (PAI-1)



Name of the stove –Dura ICS with Pot Skirt

Stove manufacturer – EcoZoom/SSM

Stove Type – Rocket

Stove Body – Cast Iron

Stove height – 32.1 cm

Stove diameter – 24 cm

Thermal efficiency⁷ – 44.9%

⁷ Thermal Efficiency was certified by the CREEC Lab, recognized certifying agency.

Stove lifetime – 7 years⁸

Distribution plan for upcoming verification(s)/instance(s):

The distribution of ICS will occur at centralized distribution points, where households from surrounding villages will be educated on the proper use and maintenance of the technologies. Shortly following the initial distribution, CHWs will visit each household that received a technology to deliver the training again, ensuring that all members of the household are aware of proper use and maintenance.

The ICS distribution is planned in close coordination with the local government offices, e.g. District Offices, Sector Offices, and so forth. There is a structured plan through which the distribution activity is scheduled and then implemented on ground.

Project activity description (including the technologies or measures employed) and how it/they will achieve net GHG emission reductions or removals.

The project activity included in this project description is the distribution and installation of high efficiency ICS systems in PAIs within Rwanda.

The ICS units will be based on a “rocket stove” design which reduce GHG emissions from biomass burning through improved combustion efficiency of wood fuel and decreased wood fuel consumption. The project stoves improve heat transfer to cooking pots through the insulated walls and design of the stove top when compared to traditional cooking methods over open- and three-stone fires. The project stoves directly decrease wood fuel demand and reduce the pressure from wood cutting within the project boundary. A 30-50% saving of firewood when compared to three-stone fire cooking has been reported in certain models. The time spent foraging for wood fuel is thereby also reduced, saving time for especially women to engage in other economic driven activities.

ICS technologies planned under the GP for distribution:

The following ICS models are envisaged to be distributed in the initial PAIs –

Model-1:

Name of the stove – Kuniokoa Generation 2 with Pot Skirt

⁸ This is based on the KPT survey experience of the PP for the Group-1 & Group-2 Dura ICS being used on-site. The manufacturer specifications only provided a range to indicate the ICS lifetime, however, the real-world experience of the PP with the monitoring of these ICS affirms its belief that these ICS will achieve the upper limit of the lifetime. Adding to the that, with the dropout factor being monitored, this should not be a concern as to whether the stoves are actually aligning with the manufacturer claimed age or not.

Stove manufacturer – Burn

Stove Type – Natural Draft

Stove Body – Stainless Steel

Stove height – 32.1 cm

Stove diameter – 28.2 cm

Thermal efficiency – 51.3%

High Power Output – 2.9 KW

Stove lifetime – 10 years

Figure 2: Kuniokoa Stove Model reference picture



Model-2:

Name of the stove – Dura 32cm With Pot Skirt

Stove manufacturer – SSM

Stove Type – Rocket Stove

Stove Body – Cold rolled steel

Stove height – 32 cm

Stove diameter – 22.8 cm

Thermal efficiency – 44.9%

Stove lifetime – 7 years

Figure 3: Dura Stove Model reference picture⁹

⁹ The picture is representative of the actual product. The design parameters are likely to remain the same as provided in the reference image, with the stove being a rocket type design. However, the color of the ICS, the DelAgua logo, and the barcode provided to capture the unique ID information of the project ICS may vary for the distribution batches. It is beyond PP's control



Model-3:

Name of the stove – S32-X With Pot Skirt

Stove manufacturer – SSM

Stove Type – Natural draft

Stove Body – Stainless Steel

Stove height – 27 cm

Stove diameter – 32 cm

Thermal efficiency – 50.1%

Stove lifetime – 10 years

Figure 4: S32-X Stove Model reference picture



List of main manufacturing/production technologies arrangements, systems and equipment involved.

Detailed application and information on the ICS technology that is going to be distributed under the new PAIs will be described in the upcoming MRs, elaborating the information under

as manufacturers may change the color or the external body in such a manner that the logo or the barcode might need to be rearranged.

relevant section of the MR for the newly included PAIs. PP will source the best available technologies for distribution under the GP. The models of the ICS technologies briefed above are indicative until the actual distribution takes place at the project location. The ICS model selection at PAI level will be based on key attributes such as thermal efficiency, durability, cultural acceptability, and manufacturing quality and warranty. From time to time, requirement gathering will be done on a continuous basis, seeking comments from the end users (through monitoring surveys, feedback requests on call center, feedback via register available at different sites, and anonymous feedback through CHWs), and based on these user feedbacks, newer models will be introduced to the GP, and new PAIs will be designed to accommodate for their variability in terms of design, specifications, and geographies within the host country (wherever required).

It will be verified by the PP that the project technologies are, at a minimum, meeting the technical specifications and requirements set in the respective methodologies (under the eligibility criteria).

At present, DelAgua is sourcing the ICS from different manufacturers, deploying them at its central warehouse through imports, and then distributing them in a scheduled manner. Each distribution involves various steps, which are elaborated in the relevant section of this document. In this whole process, quality of the equipment being distributed will be checked on spot basis. Each lot of the distribution batch will undergo spot checks, as it is an integral part of Quality Assurance at DelAgua. The ICS units that will be found as faulty, will not be distributed to the end users. At present, only rocket stove models are planned for the upcoming distribution PAIs, as their distribution is found both beneficial to the locals (usage and maintenance) as well as a successful one from the implementation point.

The types and levels of service

Level of Service: PAIs will provide ICS based on the ‘rocket stove’ design, delivering a level of service at least equivalent to the baseline appliance (open fire used for household cooking).

Type of Service: The ICS technology will deliver a specified high-power thermal efficiency of at least 25% and evidenced by a manufacturer’s specification and WBT certificate. This is a significant improvement on the baseline open fire thermal efficiency default of 10%.

List of facilities, systems and equipment in operation under the existing scenario prior to the implementation of the project.

Under the baseline scenario, households use non-renewable woody biomass for their fuel needs for cooking. The baseline appliance being replaced by the project is an open fire (typically based on the three-stone concept).

1.12 Project Location

For each PAI, the project boundary will be defined as being within the geographic borders of Rwanda. The project boundary for each Project Activity Instance is the physical, geographical location of the ICS distributed under it within the geographical boundaries of Rwanda.

Also, each individual distribution under the project technology will record the geographical location related data (e.g. latitude/longitude data), which can be verified from the recorded data (monitoring surveys).

KML files are being provided as supporting documents to demonstrate the extent, the geographical coverage of the PAIs within the GP. All the KML files will indicate the regions within the geographical boundaries of the host country Rwanda.

1.13 Conditions Prior to Project Initiation

The baseline scenario is the same as the conditions existing prior to the Grouped Project initiation.

Refer to Section 3.4 (Baseline Scenario).

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The GP complies with applicable local laws and regulatory frameworks. Under Rwandan law and regulation, none of the activities or processes relating to the manufacture or distribution of Improved cookstoves require Environmental Impact Assessments not meet any of the screening criteria in Appendix 2, page 40 of the EIA guidelines¹⁰

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

Every project activity instance that is being added to this Grouped Project, will be required to demonstrate whether they have been registered, or are seeking registration, under any other greenhouse gas programme.

¹⁰ General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda, November 2006 (See filename: RW_EIA_Guidelines_Final_versionI_Nov_2006)

Where the project activity instance has been registered under any other greenhouse gas programme, the registration number and details will be provided.

In the GP, ICS under PAI-1 were distributed in 2020 under the CDM PoA 9626 CPA-18 to CPA-26. The GHG Emission rights belonging to those ICS are transferred by the CDM PoA CME to DelAgua (DelAgua Health Rwanda Voluntary Limited), from the effective date 01-Jan-2021. The main reason for CME transferring the GHG emission rights was its inability to operate the CPAs owing to the lack of funds and interest among the financing entities. Since DelAgua was able to operate the same under VCS, it moved forward with this approach post consultation with VCS.

Also, it is to be noted that the PP (DelAgua) has no mandate or legal capacity to influence the CDM PoA 9626, for the procedures laid in any or all of its CPAs. The CDM PoA 9626 CPAs 18-26 will not undergo any issuance from 01-January-2021 or later. The CPA implementers for the CPA 18-26 are different from those for CPA 1-17, and are not related to the PP (DelAgua). Therefore, it can be ascertained that the ICS units seeking VCUs under the planned VCS Grouped Project Activity PAI-1 no longer belong to CDM, effective from 01-January-2021.

In this context, a declaration signed by the PP (DelAgua) will be shared with the VVB, stating that it has no rights to communicate with CDM under any capacity, as far as CDM PoA 9626 verifications pertaining to CPA-18 to 26 (from 01-January-2021) are concerned.

1.15.2 Projects Rejected by Other GHG Programs

Not applicable, as this project is a new Grouped Project activity and has not been rejected by other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The current information is in-line with the section and PD&MR template specific requirements for listing the PD&MR.

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

1.16.2 Other Forms of Environmental Credit

The current information is in-line with the section and PD&MR template specific requirements for listing the PD&MR.

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

Following sustainable contributions were part of the PAI – 1 (Distribution of ICS in Rwanda):

1. The proposed project activity will reduce the indoor air pollution levels to a significant level, thereby reducing the risks people face with the 3 stone fires or the traditional cooking devices. The number of deaths per year accounted to the indoor household air pollution is high in Rwanda, and lack of quality health interventions at local level increase the risk further. Lower respiratory infection is the second leading cause of human lives lost in Rwanda¹¹. Household air pollution accounts for more than 80% of morbidity damages in Rwanda¹². The lower respiratory disease risk and improved indoor air quality will reduce the mortality rates caused due to it.
2. Prior to the usage of ICS, women in Rwanda had to spend an average of 3 hours every day for collection of firewood, and this has come down significantly with the usage of project ICS. The less amount of fuel used for cooking means that they must spend less time in the field collecting wood. This time saved by the women can now be used towards better management of household, or can be used towards any livelihood or income generating activity, e.g. farming, poultry, animal products, etc. Adding to that, children too had to suffer because of this, and had to spend around 2-3 hours every day to support their families in collection of cooking fuel, i.e. firewood. The project ICS will reduce the burden on the children too, as they now will have more time to utilize. With reduced time spent on collecting firewood, they will be able to contribute more towards their education and extra-curricular activities.
3. The proposed project activity aims to distribute similar ICS units to the Rwandan households without any cost, and this will reduce their reliance on the firewood fuel usage. The project ICS models claim to reduce fuel usage by more than 50% than their baseline consumption. This will save an estimated 8-9 kg of firewood fuel usage per household on daily basis. The modern, improved cooking device will reduce the firewood usage of beneficiary households, which consequently will allow them to lead a healthy and less worried life.

¹¹ <https://www.healthdata.org/rwanda>

¹² <https://www.thelancet.com/action/showPdf?pii=S2542-5196%2821%2900201-1>

4. The proposed VCS project activity employs Community Health Workers (CHWs)¹³ on a contractual basis and provides them with the per diem for their services. These community health workers are the most important element of the GP, and therefore, the project activity will seek the services of more and more such CHWs in its lifetime. The project activity has a permanent staff, working in the central office in Kigali, and the CHWs as well as permanent staff are going to be increased in coming days, to meet its goal of distributing ICS to households in Rwanda. The CHWs are trained from time to time, in order to meet the requirements of the monitoring data recorded at the time of project ICS distribution.
5. The GHG emissions caused by the burning of firewood will get reduced by the usage of project ICS. The reduction in the firewood consumption compared with the baseline levels provide the amount of GHGs avoided, by application of VCS methodology VMR0006, ver. 1.1.

The proposed project activity will reduce the GHG emissions through the distribution of ICS technologies, and this will reduce the dependence of Rwandan people on such technologies and practices that were time, labour, and resource intensive in terms of their net output. In comparison, project technologies will provide them with cleaner alternatives, that are affordable, promotes healthy life, and are easy to operate and maintain. These interventions will contribute towards the SDG commitments of the host country Rwanda. The details on the relevant SDGs, and the impacts in terms of their net change due to the project activity will be monitored throughout the crediting period and will be reported in the monitoring reports.

The SDG impacts due to the distribution activities and monitoring related discussion is provided in the below sections.

1.17.2 Sustainable Development Contributions Activity Monitoring

The SDG impacts from the retroactive GP, and the plan for the monitoring those impacts is described in the PD&MR template table below.

¹³ CHWs are the people who are engaged with the district government supported community level health programs within the different villages in Rwanda. These CHWs are recognized by the local people in Rwanda as the support staff for village households in their health issues, and always stay connected with the local households.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	3.9	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	The proportion of the project population who feel there is a reduction in the household pollution levels will be monitored during the surveys. The survey questionnaire has a question¹⁴ specific to the variations observed by the end-user (or family member) in the indoor air pollution levels from the moment they're using the project ICS (compared with the baseline cooking device levels). This was recorded for PAI-1 ICS users. For the current monitoring period, 98.31% of respondents for PAI-1 admitted that they have experienced a reduction in the household pollution levels ever since they started using the project ICS.	As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.

¹⁴ Please refer Question with code VS82: "How does the amount of smoke generated by the Dura stove compare to the amount of smoke created by the stove you used before Dura?"

2)	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	Implemented activities to decrease	During the current monitoring period, following was observed in terms of reduction in time spent on fuelwood. PAI-1: A total of XX¹⁵ users out of XXX (average of wet & dry season) accepted that their time spent in collecting fuelwood has decreased.	As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.
3)	7.1	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Implemented activities to increase	The indicator showing the increase in the “proportion of population with primary reliance on clean fuels and technology” will be the number of users who switched to cleaner technology, i.e. project ICS in this case. The parameter will be the number of beneficiaries for a monitoring period. The value will be sourced from the ICS distribution database. During the current monitoring period, the number of beneficiaries who received the project ICS are 88,490 for PAI-1.	As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.

¹⁵ To be revised at the time of validation.

4)	8.3	8.3.1 Proportion of informal employment in total employment, by sector and sex	Implemented activities to increase	The project activity has a database of CHWs working for it. This database will be used to provide the actual number of CHWs working for the project activity. The number of permanent employees working on the DelAgua payroll will not act as a part of this indicator since it is only focused toward the informal employment. However, the number of estimated people who worked in the manufacturing of these ICS, and those working for the delivery and transportation of these ICS will be considered towards the indicator. Even if the PP is not directly employing the ironsmiths or technicians working on contract at the manufacturer's facility, purchasing of these stoves will become an indirect source of their income. Actual impact was not monitored during current monitoring period¹⁶.	As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.
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¹⁶ During the current monitoring period, the relevant Section on SDG came into effect from a time period later than the monitoring activity that was performed on-site. Due to the same, actual and estimated impacts by the proposed project activity in terms of the SDG indicators could not be monitored/assessed. The PP will ensure that the relevant section on SDGs in the VCS MR will be completed with all the information required to be filled as per the VCS MR template requirements.

5)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	The amount of GHG avoided by the project activity would be considered as final value towards this SDG indicator parameter. During the current monitoring period, the GHG emission reductions total to 683,373 tCO_{2e}.	As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.
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1.18 Additional Information Relevant to the Project

Leakage Management

Every project activity instance to be added to this Grouped Project will demonstrate that it has applied the leakage management conditions required by the respective methodologies if these are relevant to the project activity instance. If leakage is not relevant to the project activity instance, then it shall be justified in the project description.

For further information on the application of the leakage factor, please refer Section 4.3 of this document.

Commercially Sensitive Information

All information in this Grouped Project Description is public.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

The project undertakes the distribution of improved cook stoves. As per the Laws of Rwanda, the project undertaking such activities does not require any Environmental Impact assessment. This is as per Ministerial Order No. 004/2008: Ministerial Order establishing the list of works, activities and projects that must undertake an environment impact assessment.¹⁷ The fact that this project activity does not need environmental assessments, supports that it does not cause environmental harm.

The project, on the contrary, has several positive social and environmental benefits due to its implementation. The project leads to improved indoor air quality, lower air emissions, and also positive health impact.

2.2 Local Stakeholder Consultation

Since this is a retroactive project activity, the LSC related requirements on the Local Stakeholder Consultation meeting was fulfilled at the time of registration of CDM PoA. This

¹⁷ <https://www.elaw.org/rwanda-environmental-impact-assessment-law-and-regulations>

was done through a nationwide LSC meeting conducted prior to the PoA implementation. The LSC requirements were duly validated by a third-party auditor (DOE) for CDM PoA validation, and was further reviewed by CDM team. The documents are publicly available on the PoA webpage¹⁸ on CDM registry.

As far as continuous grievance of the local stakeholders is concerned, the required measures are already in place to account for a grievance through any media. These will be verified by the VVB at the time of validation, and shall remain open for Verra for review.

2.3 Environmental Impact

The project undertakes the distribution of improved cook stoves. As per the Laws of Rwanda, the project undertaking such activities does not require any Environmental Impact assessment. This is as per Ministerial Order No. 004/2008: Ministerial Order establishing the list of works, activities and projects that have to undertake an environment impact assessment¹⁹.

2.4 Public Comments

The public commenting period is pending for the GP, and relevant comments raised during the public comment period will be added to this section (alongwith the responses that were provided against them) at the time of validation.

2.5 AFOLU-Specific Safeguards

Not applicable since the GP is not an AFOLU activity.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Grouped Project activity will involve PAIs that will apply the following methodology, approved under VERRA:

¹⁸ https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/RVUJL2DMZ7HBKC1PEG4O8NAOWSI6TX/view?cp=1

¹⁹ <https://www.elaw.org/rwanda-environmental-impact-assessment-law-and-regulations;>

Specific URL to the Order No. 004/2008:

<https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Felaw.org%2Fsystem%2Ffiles%2Fattachments%2Fpublicresource%2FMINISTERIAL%2520ORDER%2520N%25C2%25B0004%2520ESTABLISHING%2520THE%2520LIST%2520OF%2520WORKS.docx&wdOrigin=BROWSELINK>

- VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves – version 1.1

In the context of applied methodology, the associated tools and guideline documents applicable for the Grouped Project include:

- CDM TOOL01 “Tool for the demonstration and assessment of additionality” Version 07;
- CDM TOOL21 “Demonstration of additionality of small-scale project activities” Version 13.1;
- CDM TOOL30 “Calculation of the fraction of non-renewable biomass” Version 03;
- CDM Guideline “Sampling and surveys of CDM project activities and programmes of activities” version 04;
- CDM Standard “Sampling and surveys for CDM project activities and programmes of activities” version 09.

Project Activity Instance 1 is applying the following approved methodology:

- VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves – version 1.1

3.2 Applicability of Methodology

The applicability criteria of the methodologies and tools used are justified at the project activity level. This is reflected in the following tables in this section.

a) Below is the methodological applicability criteria for the PAIs that are going to be included in this VCS grouped Project Activity.

Table 5: Fulfilment of VMR0006, ver. 1.1 Applicability criteria for PAIs to be included

S. No.	Applicability Condition	Justification of applicability
1	Project activities shall be implemented in domestic premises or in community-based kitchens.	Each PAI will distribute technology that is appropriate for domestic use to household end users
2	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's	ICS distributed in each PAI will have a thermal efficiency of at least 25%

	specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	Each PAI will record the baseline and project fuel use to ensure that wood fuel is the sole source of energy.
3	Both 'Projects' and 'Large Projects' can use the methodology	PAIs will comply with the GP requirements
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Each PAI will demonstrate that Non-renewable biomass has been used in the project region since 31 December 1989.
5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Biomass residues are not processed as a fuel to be used in the GP.
6	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	In order to avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient

		Each PI will have system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including: 4. Date of distribution 5. A unique serial number on each technology Available end-user data: name, phone number, address/physical location, ID number
7	The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The End User Agreement confirms that the ownership rights of the ER/VCUs generated by the PAI reside with the PP. The signature of each end user confirming their understanding of this agreement will be collected at the point of distribution. The End User Agreement will be in English and translated into a local language where required. If necessary, it will be read out to the End User. All rights to distribute the delivered stoves and claim the resultant emissions reductions belong to the PP. The GP will not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.

b) Below is the table showing fulfilment of methodological applicability criteria for the PAI that is undergoing monitoring in the VCS grouped Project Activity, i.e. PAI-1.

Table 6: Fulfilment of VMR0006, ver. 1.1 Applicability criteria for PAI-1

S. No.	Applicability Condition	Justification of applicability
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1	Project activities shall be implemented in domestic premises or in community-based kitchens.	Under PAI-1, PP has distributed ICS to the households in Rwanda in 2020.
2	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	ICS distributed under PAI-1 have the thermal efficiency >25%, which is evident from the respective year of the monitoring applicable to the ICS life. For year 2021 of monitoring period, the thermal efficiency for the first and second year from the year/date of PAI-1 distributions were applied. Similarly, subsequent years of the monitoring period applied the applicable drop in the efficiency value. These can be verified from the ER Calculation Sheet provided alongwith the relevant version of PD&MR document. All the applied values during the first monitoring period are >25% thermal efficiency value. The woody biomass usage is monitored during the first monitoring period, at both baseline level as well as project level. This can be verified from the monitoring surveys for PAI-1.
3	Both 'Projects' and 'Large Projects' can use the methodology	PAI-1 within the GP is categorised under large project category, since the number of annual VERs is >300,00 tCO _{2e} .
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	The PAI-1 is implemented in the host country Rwanda, and it has already been mentioned in Section 1.4 by using the available literature that the host country Rwanda has been using firewood as a source of fuel for their cooking requirements.

5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	This is not applicable for the PAI-1.
6	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	In order to avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient PAI-1 was implemented with the system for avoiding double counting, with data being collected via a smartphone app, including: 1. Date of distribution 2. A unique serial number on each technology 3. Available end-user data: name, phone number, address/physical location, ID number
7	The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed	The End User Agreement confirms that the ownership rights of the ERU/VCUs

	procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	generated by the PAI-1 remain with the PP. The signature of each end user confirming their understanding of this agreement is collected at the point of distribution. The End User Agreement is bilingual, i.e. in English as well as translated into the local language Kinyarwanda. In cases where the end-user was unable to understand the agreement, it was read out to them for the sake of their clarity. All rights to claim the resultant emissions reductions for the ICS distributed under the PAI-1 belong to the PP. PAI-1 has not been rejected by another GP, or PoA-DDs of any other GHG programs.
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3.3 Project Boundary

The Grouped Project boundary is defined as per the applied methodology VMR-0006, ver. 1.1.

Table 7: The GHG sources & sinks within the project boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Major source
	CH ₄	Yes	Major source
	N ₂ O	Yes	Major source
	Other	No	No other source identified
Project	CO ₂	Yes	Major source
	CH ₄	Yes	Major source

Source	Gas	Included?	Justification/Explanation
Biomass for household cooking requirements)	N ₂ O	Yes	Major source
	Other	No	Not applicable

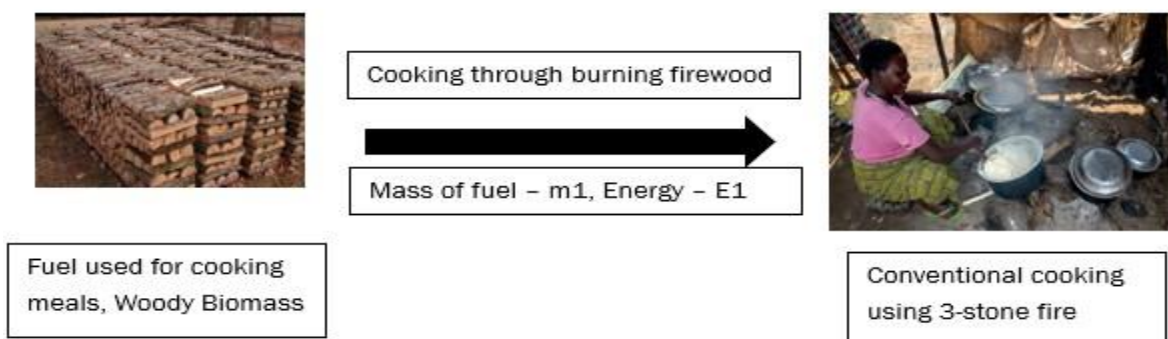
Figure 5: Project boundary map showing physical locations of the Grouped Project distribution



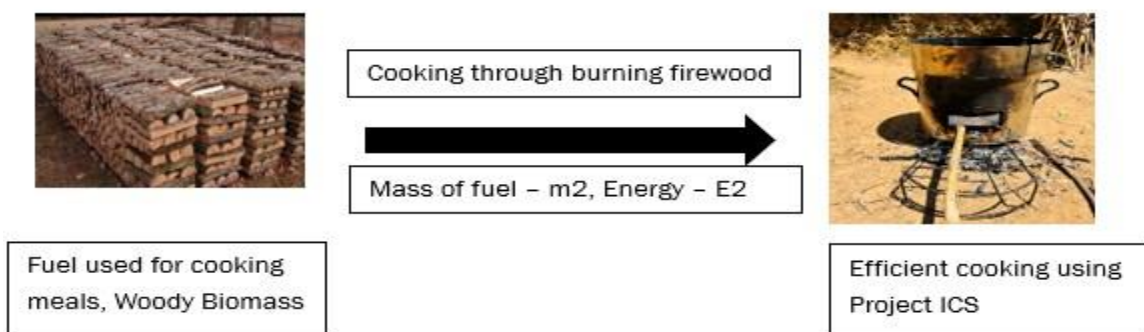
As per the template requirements, the diagram of the equipments distributed under the grouped project is provided in Section 1.

Flow of mass and energy within the grouped project activity is described below.

Baseline Scenario:



Project Scenario:



Here, the change in mass of firewood between baseline and project scenario is $m_1 > m_2$.

The change in energy consumed for cooking between baseline and project scenario is $E_1 < E_2$.

The systems that were applied for the distribution activity to take place, are described in the Section 5.3 of this document.

3.4 Baseline Scenario

The baseline scenario has been identified as the continuation of the current practice of the preparation of meals, using conventional inefficient stoves. In rural areas, the reliance on biomass is over 90%. Most Rwandans live in rural areas where traditional biomass, mainly wood fuel has remained the leading source of energy for cooking²⁰. In the baseline scenario, the greenhouse gas emissions are therefore determined based on the amount of energy used by the cooking fuel used in the conventional stove application. The baseline scenarios

²⁰ <http://www.reg.rw/what-we-do/biomass/>

before the implementation of this Grouped Project is the use of non-renewable biomass, mainly firewood for cooking in households throughout the project boundary.

Since the grouped project activity provides improved cookstoves to households where there was only firewood was used as the cooking fuel, and the cooking device was either a traditional cooking device (e.g. rondreza) or 3 stone fire, it is considering the burning of firewood, and usage of unimproved cooking devices as the baseline. Also, PP verifies this for every individual household where distribution of project ICS is taking place, that they were using unimproved cooking device using firewood as fuel prior to receiving it. This is acknowledged by the end-user in the form of a signed ICS usage practices poster where the factual information is written down. This information can be checked through the sample posters. The poster is in local language Kinyarwanda, which makes it clearly understandable for the end user.

Apart from the above, there are several documentary evidences in the form of recently published reports and papers that can support the fact that the project's baseline was usage of firewood and unimproved cooking devices.

Some of these are listed below:

1. Ministry of Infrastructure: Biomass accounts for 85% of all energy consumed²¹. Under the National strategy for transformation, the sector objective is to halve the number of households using traditional cooking technologies to achieve a sustainable balance between supply and demand of biomass through promotion of most energy efficient technologies.
2. Rwanda Integrated Household Living Conditions Survey, EICV5: In rural areas, firewood remains the most common type of cooking fuel, used by 93% of the households²².

Rwanda first Biennial Report to UNFCCC: Biomass (mostly wood fuel), is the most used fuel in households for cooking (83 % of households in 2019) and as a source of energy (86.3 % in 2016/2017)²³.

3.5 Additionality

The Additionality for Improved Cookstoves is demonstrated using the requirements provided in the methodology VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves – version 1.1

²¹ <https://www.mininfra.gov.rw/digital-transformation-1-1>

²² <https://www.statistics.gov.rw/publication/eicv-5-main-indicators-report-201617>

²³ https://unfccc.int/sites/default/files/resource/Rwanda%20First%20Biennial%20Update%20Report_Final_V.pdf

As per the Para-7, step-2 of the methodology VMR0006, the applicability conditions described within the methodology represent the positive list. The project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition. In so doing, the project is deemed as complying with the positive list. The project's compliance to applicability conditions has been demonstrated in section 3.2 above. In addition, the project meets the below condition:

- Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

In case of this VCS grouped project activity, all the improved cookstoves are being distributed at zero cost to the end user. Hence the grouped project activity is deemed additional.

Hence, as discussed above, distribution of Improved cook stoves within the boundaries of Rwanda may be established as additional.

Therefore, the Grouped project activity is deemed as additional.

3.6 Methodology Deviations

Not applicable.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The applied methodology VMR0006, ver. 1.1 does not account for baseline emissions separately, rather quantifies emission reductions as a function of the reduction in the amount of non- renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. See section 4.4 below.

4.2 Project Emissions

The applied methodology VMR0006, ver. 1.1 does not account for project emissions separately, rather quantifies emission reductions as a function of the reduction in the amount of non- renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. See section 4.4 below.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with section 8.3 of VMR0006.

4.4 Estimated Net GHG Emission Reductions and Removals

Equations used for Improved Cookstove as per VMR0006

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

i = Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire

j = Indices for the situation where there is more than one batch of improved cook stove of type i

ER_y = Emission reductions during year y in t CO₂e

$ER_{y,i,j}$ = Emission reductions by improved cook stove of type i and batch j during year y in t CO₂e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf\ CO_2} + EF_{wf\ non-CO_2}) \times N_{y,i,j} \times 0.95$$

Equation (2)

Where:

$B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y

$f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (fNRB)²⁴

$N_{woodfuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

²⁴ Default values endorsed by designated national authorities and approved by the Board are available at <https://cdm.unfccc.int/DNA/fNRB/index.html>

$EF_{wf\ CO_2}$ = CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)

$EF_{wf\ non-CO_2}$ = Non-CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO₂/TJ)

$N_{y,i,j}$ = Number of improved cook stoves of type i and batch j operating during year y

0.95 = Discount factor to account for leakage

The quantify of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves will be estimated as follows:

$$B_{y,savings,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}} \right) \quad \text{Equation (3)}$$

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (4)}$$

Where:

B_{old} = Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of B_{old} has been sourced from baseline field surveys.

η_{old} = Efficiency of baseline cookstove

$\eta_{new,i,j}$ = Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT).

Efficiency of the improved cookstove will be updated as per equation below:

$$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (5)}$$

Where:

η_p = Efficiency of project stove (fraction) at the start of project activity

DF_n^{y-1} Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available

literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

If project households continue to use baseline cookstoves along with improved cookstoves, B_{old} shall be adjusted ex-post based on the percentage of project households found to continue such practice according to Equation 6 below. For such cases, the quantity of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove.

$$B_{old,adjusted} = B_{old} \times (1 - \mu_y) \quad \text{Equation (6)}$$

Where:

$B_{old,adjusted}$ = Adjusted B_{old} to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device)

μ_y = Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves.

Sample VCU estimates (based on the calculation for first year of ICS monitoring estimates):

If the baseline stove usage is 10.23% of the total cooking in the first year, then the factor μ_y will be 0.1023.

And based on that,

$$B_{old,adjusted} = 4.409 \times (1 - 0.1023) \text{ tonnes/year}$$

$$\text{Or } B_{old,adjusted} = 3.958 \text{ tonnes/year}$$

$$B_{y,savings,i,j}^{25} = 3.958 \times (1 - (0.1/0.4221)) \text{ ton firewood/household/annum}$$

$$\text{Or } B_{y,savings,i,j} = 2.998 \text{ ton firewood/household/annum}$$

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf \text{ CO}_2} + EF_{wf \text{ non-CO}_2}) \times N_{y,i,j} \times 0.95$$

$$ER_{y,i,j} = 2.998 \times 0.9682^{26} \times 0.0156 \times (112+26.23) \times 1 \times 0.95$$

²⁵ Assuming here that all the ICS distributed were of Kuniokoa model (bearing 51% thermal efficiency), and there was no loss in efficiency due to the non-usage of pot skirt.

²⁶fNRB value approved for host country Rwanda under GS VPA 11133; URL: <https://registry.goldstandard.org/projects/details/3088> Further details are provided in the Section 5.1 table f_{NRB} of this PD&MR.

$ER_{y,i,j} = 4.59 \text{ ton CO}_2/\text{annum}^{27}$.

Detailed calculations on the VERs, for both (i) estimates throughout the crediting period, and (ii) actual VERs obtained in the first monitoring period, are provided in the ER Sheet.

Emission Reduction Estimate for Improved Cookstove²⁸

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
01/01/2021 – 31/12/2021	404,784	-	-	404,784
01/01/2022 – 31/12/2022	391,761	-	-	391,761
01/01/2023 – 31/12/2023	376,153	-	-	376,153
01/01/2024 – 31/12/2024	365,568	-	-	365,568
01/01/2025 – 31/12/2025	349,451	-	-	349,451
01/01/2026 – 31/12/2026	338,122	-	-	338,122
01/01/2027 – 31/12/2027	71,568	-	-	71,568
Total	2,297,407	-	-	2,297,407

The VCS methodology, VMR0006 is applicable to both 'Projects' and 'Large Projects'. Hence there are no limits on volume of credits from Improved Cookstove component that can be certified per annum.

5 MONITORING

5.1 Data and Parameters Available at Validation

²⁷ The sample calculations are for Kuniokoa ICS distribution, and ERs are calculated for its first year of operations. In the sample calculations, the accounting of discount factors such as drop-off factor and the skirt usage rate were not accounted.

²⁸ For the crediting period estimates, due consideration was given to the discount factors, thus using realistic figures for the project scenario.

Data/parameter fixed ex-ante for Improved Cookstove instances

Data / Parameter	B_{old}
Data unit	tons/annum
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices
Source of data	Baseline KPTs, conducted according to KPT Version 4.0
Value applied:	4.4092 ²⁹
Justification of choice of data or description of measurement methods and procedures applied	The sampling plan utilized multi-stage sampling. The primary sampling unit was rural villages in Northern, Southern, Eastern, Western Provinces, and the secondary sampling unit was households. 5 households were randomly selected from 25 randomly selected villages. All KPTs were conducted by a team trained by Berkeley Air Monitoring Group. The results presented above met the required 90/10 confidence/precision. The full dataset and analysis have been provided to the VVB.
Purpose of Data	Calculation of emission reductions
Comments	B_{old} will remain fixed for the entire crediting period.

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity

²⁹ A baseline firewood consumption of 12.08 kg/HH/day converted into tons/hh/annum. Calculations may be checked in the ER Sheet.

Source of data	Certification by the Centre for Research in Energy and Energy Conservation (CREEC) lab. CREEC is a certifying agent recognized by the Rwandan Standards Bureau (RSB).
Value applied:	Kuniokoa Generation 2 0.4387 for project stove without skirting³⁰ 0.51 for project stove with pot skirting³¹ Dura Woodstove 0.449 for project stove with skirting³² 0.409 for project stove without pot skirting³³
Justification of choice of data or description of measurement methods and procedures applied	This data is fixed ex-ante as per applied methodology VMR0006.
Purpose of Data	Calculation of emission reductions
Comments	If a pot-skirt is distributed with the project stove, the parameter will be calculated as a weighted average of the efficiency of a skirted stove and an unskirted stove.

Data / Parameter	Life Span
Data unit	Years
Description	State the operating lifetime of project device for projects opting Equation 6 (above) for updating project stove efficiency during project crediting period.
Source of data	Manufacturer's specifications
Value applied:	Kuniokoa Generation 2 10 years

³⁰ Certification report provided to the VVB

³¹ Certification report provided to the VVB

³² Certification report provided to the VVB

³³ Certification report provided to the VVB

	Dura Woodstove 7 years
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of emission reductions
Comments	PP is considering the upper limit of the project ICS model Dura lifetime, as the lifetime of the Dura provided under the technical specifications is 6-7-year range. The reason for considering the upper limit of the ICS lifetime range for this GP is the PP's experience with the actual usage of the ICS model on-site. During the first monitoring period surveys (in late 2021), it has been found that a significant number of project ICS are still working, and people are using them for cooking. Discarding/abandoning the ICS for ER calculations, which are in use in the 7 th year of their life does not seem viable for a project activity which is solely relying on the carbon finance for its sustenance, and the livelihoods of the employees working under the project activity. Using the lower end of the lifespan may have negative implications for the program's revenue stream.

Data / Parameter	NCV _{biomass}
Data unit	TJ/ton
Description	Net calorific value of the non-renewable woody biomass
Source of data	IPCC
Value applied:	0.0156 TJ/tonne
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$EF_{wt\ CO_2}$
Data unit	ton CO ₂ /TJ
Description	CO ₂ emission factor for woody biomass
Source of data	IPCC
Value applied:	112 tCO ₂ /TJ
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$EF_{wt\ non-CO_2}$
Data unit	ton CO ₂ /TJ
Description	Non-CO ₂ emission factor for woody biomass
Source of data	IPCC
Value applied:	26.23 tCO ₂ /TJ
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	Leakage Factor
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Data unit	Fraction
Description	net to gross adjustment factor to account for the leakage
Source of data	Default as per VMR0006, ver. 1.1, Section 8.3
Value applied:	0.95
Justification of choice of data or description of measurement methods and procedures applied	In-line with the applied methodology VMR0006, ver. 1.1 Section 8.3.
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$f_{NRB,Y}$
Data unit	Fraction
Description	Fraction of woody biomass that can be established as non-renewable biomass
Source of data	Gold Standard ³⁴ PoA 10959 VPA – 02: “SAFE WATER PROJECT IN RWANDA II ³⁵ ”, GSID - 11133
Value applied	0.9682
Justification of choice of data or description of measurement methods and procedures applied	The applied value is taken from a publicly available GS VPA, where the f_{NRB} value is calculated by the CME <i>for the host country Rwanda</i> , and the relevant section of the <i>validation report</i> (Section D.8.1, Page 32) assessment states that it is calculated by applying the latest available version of CDM Tool-30 (version 03.0) and has been validated by an approved VVB ³⁶ .

³⁴

³⁵ <https://registry.goldstandard.org/projects/details/3088>

³⁶ The validation report for the validated f_{NRB} value is being shared with the VVB.

Purpose of Data	Calculation of emission reductions
Comments	<i>The applied f_{NRB} value has been calculated by using the latest available data sources (as mentioned in the VPA-DD & the VPA Validation Report), and since it is published on the GS VPA 11133 webpage, it infers that this f_{NRB} value is approved by the GS Board.</i>

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring surveys
Description of measurement methods and procedures applied	Measured directly from Project database. A discount shall be applied based on the percentage of devices that are operational during the monitoring period, as determined by the sample survey. For instance, if survey results show that 10% of the devices are nonoperating, an adjustment factor of 0.9 shall be applied to number of project devices commissioned in a particular batch. Separate samples shall be taken for each batch. Simple random sampling approach will be used in sampling surveys. Minimum sample size will be determined as per the following guidelines: – Project target population < 300: Minimum sample size 30 – Project target population 300 – 1000: Minimum sample size 10% of group size Project target population > 1000: Minimum sample size 100
Frequency of monitoring/recording	Atleast once every two years

Value applied	PAI – 1: Actual distribution of 88,490 ICS.
Monitoring equipment	Surveys
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	This parameter only accounts for the number of project ICS that are operating in the monitoring period, and rest of the factors such as discount on non-usage of project ICS, usage of pre-project device, and drop-off are accounted separately, and hence are not reflected with the values provided for this parameter. Here, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	η_{old}
Data unit	Fraction
Description	<i>Efficiency of baseline stove</i>
Source of data	Calculated using the Default value as specified in VMR0006 methodology and the data obtained through the distribution surveys
Description of measurement methods and procedures applied	If the device is a three stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney, a default value of 0.1 is used as per the applied methodology; for other types of devices, a default value of 0.2 is used.
Frequency of monitoring/recording	Fixed for each individual household at first HH visit
Value applied:	0.1

Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	The value is recorded at the time of distribution, through the distribution survey. If the baseline cooking device is traditional device (e.g. Rondereza) or a 3 stone fire, without any grate or chimney, then as per VMR0006, ver. 1.1, the applied value will be 0.1. In case there is any other unimproved cooking device identified as baseline cooking device, then the applied value for the parameter will be 0.2. The average of the η_{old} recorded for total project ICS households (in the distribution surveys) will be calculated for VCU calculations. For ex-ante estimates, the applied value is 0.1 (assumption for the ex-ante calculations for crediting period-1), which will vary for each monitoring period, based on the actual data obtained at the time of distribution.

Data / Parameter	$\eta_{new,i,j}$
Data unit	Fraction
Description	Efficiency of the device of each type i and batch j
Source of data	Calculated as per the equation 5 of the applied methodology
Description of measurement methods and procedures applied	Efficiency of the improved cookstoves will be estimated each year using equation (5) above where loss in efficiency per year is calculated, and therefore no direct measurement is involved.
Frequency of monitoring/recording	Annually

Value applied:	ICS efficiency value for instances where pot-skirt is being used:		
	Crediting Period Year	Project Technology Efficiency value	
		Dura	Kuniokoa
	Year-1	0.4263	0.4794
	Year-2	0.4178	0.4746
	Year-3	0.4137	0.4699
	Year-4	0.4095	0.4652
	Year-5	0.4054	0.4605
	Year-6	0.4014	0.4559
	Year-7	0.3974	0.4513
	ICS efficiency value for instances where pot-skirt is not being used:		
	Crediting Period Year	Project Technology Efficiency value	
		Dura	Kuniokoa
	Year-1	0.3845	0.4124
	Year-2	0.3806	0.4082
	Year-3	0.3768	0.4042
	Year-4	0.3730	0.4001
	Year-5	0.3693	0.3961
	Year-6	0.3656	0.3922
	Year-7	0.3620	0.3882
Monitoring equipment	N/A		

QA/QC procedures applied	For PAI-1, only Dura ICS was distributed. Therefore, only Dura ICS thermal efficiency is considered for the first monitoring period.
Purpose of data	Calculation of emission reductions
Calculation method	As per equation (6) above
Comments	If a pot-skirt is distributed with the project stove, the parameter will be calculated as a weighted average of the efficiency of a skirted stove and an unskirted stove. Surveys shall be conducted to determine the number of cooking events that occur with and without a pot skirt. Here, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Monitoring
Description of measurement methods and procedures applied	If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting evidences to determine the frequency of usage of both the improved cookstoves and baseline cookstoves. For example, if there were 3 baseline cookstoves in a household and it was determined during the survey that use of one of them continues during the crediting period then a conservative adjustment factor of 0.33 is applied to Bold. Another example would be the case where there was only one baseline cookstove per household and its use during the project period continues along with the improved cookstove to

	meet 25% of the cooking needs of the household in which case the adjustment factor will be 0.25.
Frequency of monitoring/recording	At least once every two years
Value applied:	0.19 ³⁷
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions
Calculation method	N/A
Comments	N/A

5.3 Monitoring Plan

The monitoring plan for this project is closely derived from the methodologies. A database for the project activity will be maintained by the DelAgua.

1. Monitoring Plan Design for the Grouped Project Activity (from Monitoring Period-2 and onwards):

The monitoring plan shall consist of checking a representative sample of all ICS units distributed at least once every two years to ensure that they are still operating or are replaced by an equivalent in-service unit. A project database will be populated during the distribution of each initial cookstove issued, and will be updated based on subsequent replacements, as well as detailed data on the representative sample surveyed for monitoring purposes. The database will be accessible to the project proponent, appropriate partners, and the verification VVB. The database will include, at minimum, the following:

- Unique project identification number
- Installation date
- Family name
- Contact details of user (when available)

³⁷ Applied as assumption for year-1 estimations. Post that, a linear degradation of 2% is considered for the ER estimations.

- Socioeconomic level (if households or communities are targeted), or type of establishment (if SMEs are targeted)
- Baseline cooking fuel source
- Baseline stove type
- Date of replacement of cook stove
- Monitored parameters
- Usage of Skirt with the project ICS

The database will be available to select a random, representative sample for monitoring and verification purposes. This sample set will be integrated into the database to include additional monitoring parameters as required or as appropriate. Monitoring surveys will include examination of the cookstove installations and verification that they are in an operational condition. Units that have been damaged or aged beyond expected operational condition will be replaced. Therefore, should some appliances not be replaced appropriately, this will be accounted for and appropriately deducted from emission reduction claims.

1. Responsibility

The main roles and responsibilities of the parties involved is as described below:

DeAgua

- Acts as the project managing entity
- Development of PDD
- Registration of the project with VERRA
- Inclusion of new PAIs upon satisfaction of the eligibility criteria stipulated in the PDD
- Authorized entity for any official communication with the VERRA and VVB
- Manage and archive the database for all appliances
- QA/QC of database during distribution & data collection
- Maintain local repair/replacement facilities
- Education and public awareness raising during initial distribution as well as periodic education campaigns.
- Data collection during distribution & delivery of data to database

- QA/QC of data collection
- Conduct ongoing monitoring, as required to ascertain monitored parameters.

2. Quality Assurance and Quality Control

Spot checks of the database are conducted during the baseline surveys, distribution, and monitoring activities. Distributors are trained in the proper installation, use, and maintenance of the devices, and are provided with training sufficient to introduce recipient households to the proper operation and maintenance of the devices, as well as contact information for DelAgua. Supervisors receive similar training, as well as supervisory training in monitoring of distributor's activities. Supervisors are responsible for approximately 10 distributors. During the campaign, DelAgua staff will spot-check distributor quality, by revisiting 2 households per distributor, to verify that the stoves and /or lamps are in place, and that the recipient can show proper operation of the units. The results of this spot check are recorded electronically and compared against the initial distribution logs. The data collected in the database is verified and controlled in the following way:

1. Surveys are conducted by staff trained by DelAgua.
2. Supervisors are separately trained by DelAgua to spot-check surveyor results.
3. All ex-ante and monitored data parameters are collected on smart-phone based surveys and are uniform across all surveyors for any given survey activity.
4. Unknown to the surveyors, the database analysis includes verification of "valid" surveys including by a.) verifying household consent is received and b.) that the survey duration, as recorded by Internet time, is no shorter or longer than a reasonable value. For example, the baseline surveys for Rwanda were validated against a minimum of 20 minutes and a maximum of 120 minutes.
5. Data validation is conducted on the database to ensure that values are numerically consistent, i.e. multiple choice values add up to 100% of respondents.

3. Training and maintenance procedures:

DelAgua is responsible for training any contractors used during distribution, education and monitoring activities. DelAgua ensures training of all on-site staff with respect to adherence to the Monitoring Plan of the project activity. Records of the training will be kept. Internal audit of all the records of the distribution, monitoring and education activities are performed on an annual basis, and will be carried out once (or twice, based on the requirements and plans) a year. Training records are saved for such training programs, where step-by-step

training is provided to the surveyors for a quality data collection process, where they are expected to not impact the materially, degree of conservativeness, and perform the surveys in an unbiased manner. The surveyors are trained on a periodic basis, and refresher trainings are also provided to improve the quality of surveys from time to time. During these audits all the data and parameters that need to be monitored as per the monitoring plan will be checked and shortcomings if any will be reported and addresses. Since the surveys are electronic in format, recording of responses and the record keeping procedure for these surveys are conveyed to the team through rigorous trainings.

4. Calculation of emission reductions:

Emission reductions will be calculated using the results of the most recent survey data. The updated parameters adjust all emission reduction results for the year monitored.

5. Sampling Plan:

The sampling plan is intended to enable the discovery of unbiased and reliable estimates of monitored parameter values used in the GHG emission reduction calculations. The sampling plan design is based on the CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 4.

6. Sampling Design:

a. Objectives & Reliability Requirements

Sampling activities are defined at the Grouped Project-level and the sampling plan presented here will apply to assess parameter values in PAIs to be included in the GP, either individually or in groups.

Samples will be calculated on a 90/10 confidence/precision for individual PAIs on an annual basis, or 95/10 confidence/precision basis for groups of PAIs or sampling on a biennial basis. This is in accordance with the Para 11 of the CDM Sampling Standard, ver. 09.0.

In cases where the confidence/precision are not achieved, the lower bound of the confidence interval of the parameter value may be chosen as an alternative to repeating the survey efforts.

Parameter values may be estimated by applying a sampling in accordance with the latest version of the CDM Sampling Standard for a single PAI or a group of PAIs. A single sampling plan covering a group of PAIs is justified when either the homogeneity of included PAIs

relative to the parameters of interest can be demonstrated or the differences among the included PAIs is taken into account in the sample size calculation.

b. Target Population

The target population is the combined unique entries of ICS users by PAI in the monitoring database from which the sample is being drawn.

c. Sampling Method

For each survey, the sampling method is determined by using a random sample of the total population of distributed ICS from records generated in the monitoring database and based on the precision requirements listed above, and as required by the methodology and guidance in the CDM Sampling Standard.

Cross-PAI sampling may be carried out. In such cases, 95/10 confidence/precision will be applied.

d. Sample Size

The sample size for each monitoring activity will be the total number of ICS distributed and included in the monitoring database. A sample size calculation sheet will be used to calculate the number of samples.

Oversampling will be conducted to ensure that the required level of precision is met.

The equations used for calculation of sample size are as follows--

For proportion-based parameters:

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 p(1-p)} \quad \text{Equation (1)}$$

Where:

n	=	Sample size
N	=	Total number of households (640,000)
p	=	Our expected proportion (0.50)
1.645	=	Represents the 90% confidence required
0.1	=	Represents the 10% relative precision ($0.1 \times 0.5 = 0.05 = 5\%$ points either side of p)

For mean-based parameters:

$$n \geq \frac{1.645^2 NV}{(N-1) \times 0.1^2 + 1.645^2 V} \quad \text{Equation (18)}$$

Where:

V	=	$\left(\frac{SD}{mean} \right)^2$
n	=	Sample size
N	=	Total number of households
<i>Mean</i>	=	Our expected mean (3.5 hours)
<i>SD</i>	=	Our expected standard deviation (2.5 hours)
1.645	=	Represents the 90% confidence required

e. Sampling Frame

The Sampling Frame will be the monitoring database which may contain information from multiple PAIs and therefore allows for sampling to be applied per PAI or as a group of PAIs.

2. Monitoring Plan applied for Monitoring Period 1 (PAI-1):

Sample Size calculation -

1) Sample size calculated for the monitoring survey: During the current monitoring, sample size calculated for the PAI-1 ICS came out to be XXX³⁸. The considerations, application of equations, and other relevant information can be accessed from the sample size calculation sheet shared with this PD & MR document.

The sample size calculation for the PAI-1 returned a value of XX, which was checked with the methodological requirement of minimum 100 samples, and based on that, higher of the 2 values was applied.

The higher value than provided in the methodology is acceptable, since it is only adding to the degree of representativeness of the sample size when compared with the actual population.

For the baseline KPT, a sample size of 100 was selected as it was meeting the methodological requirements. However, accounting the non-response scenario for the

³⁸ This information shall be provided at time of validation and verification review by third party VVB.

surveys, the number of samples was set to be XXX. Since the number of people living in Rwanda is 12,955,736³⁹, it is evident that it is >1,000. Therefore, the applied sample size is meeting the applicable sample size calculation requirements⁴⁰ of the applied methodology. During the KPT surveys, a total of XXX respondents were surveyed, in accordance with the KPT protocol.

Measures for Reliability, Representativeness & Conservativeness:

1. For baseline KPTs, there was no requirement to apply the confidence/precision at the Bold value obtained, as per the applied methodology VMR0006, ver. 1.1. However, PP applied 90/10 confidence/precision only to make sure that the degree of representativeness and the reliability expected is being met for the value obtained through the surveys.
2. For the monitoring surveys, applied methodology only requires the confidence/precision to be at 90/10 while surveying individual PAIs. However, PP took it a notch higher, thus compensating for the delay in the monitoring, as it was beyond PP's control. The degree of representativeness and reliability is met for the samples and the parameters surveyed since these are fulfilling the 95/10 confidence/precision requirements.
3. There are parameters such as drop-off factor and Skirt usage rate that are monitored during the monitoring surveys, in order to arrive at the higher degree of conservativeness, thus only accounting the actual number of operational ICS on-site, and the application of reduced thermal efficiency for the usage case where the project ICS is used without skirt respectively. These are not required to be monitored by the applied methodology, however, are monitored only to account for a conservative calculation of VCUs obtained.

The number of operational days is another parameter that is monitored to account for the period of time when the project ICS was not used. This is calculated as average of the usage period (in number of days) for ICS that were found operational during the monitoring period. Rather than assuming that the ICS that were in operation operated for the whole monitoring period (all number of days within the monitoring period), PP is accounting for the number of

³⁹ <https://www.statistics.gov.rw/statistical-publications/subject/population-size-and-population-characteristics>

⁴⁰ Applied methodology VMR0006, ver. 1.1, Page 11: *Baseline Survey of Local Usage. Project proponent shall carry out a survey of usage prior to implementation of the project activity following the sampling approach described in the latest version of CDM document Sampling and surveys for CDM project activities and programme of activities. Alternatively, the project participant may follow the simple random sampling approach and the minimum sample size should be determined as per the following guidelines:*

- Project target population < 300: Minimum sample size 30
- Project target population 300 – 1000: Minimum sample size 10% of group size
- Project target population > 1000: Minimum sample size 100

days project ICS were not operating. The parameter is resulting in conservative VCU calculation for the project activity.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

The monitoring period 1 for PAI-1 covers the period from 01/01/2021 to 30/06/2022. As monitoring occurs on an annual basis, the full monitoring period is separated into the following subperiods:

Period 2021: 01/01/2021 – 31/12/2021

Period 2022: 01/01/2022 – 30/06/2022

All monitoring surveys were conducted between DD/MM/YYYY and DD/MM/YYYY⁴¹.

Data/parameter monitored for Improved Cookstove instances

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Value applied:	Improved Stoves distributed under PAI-1 = 88,490 stove units Operational rate of ICS found during M.P.-1 monitoring surveys: PAI-1 = 100% ICS considered as operational for the monitoring period – 1: 88,490>
Comments	Here, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

⁴¹ This information shall be provided at time of validation and verification review by third party VVB.

Data / Parameter	$\eta_{new,i,j}$		
Data unit	Fraction		
Description	Efficiency of the device of each type i and batch j		
Value applied:	PAI	With Pot Skirt	Without Pot Skirt
	PAI-1	41.65%	37.94%
	The calculated values obtained for PAI-1 are: PAI-1 = 0.4114 or 41.14%		
Comments	Calculated using Equation 5 of the applied VCS methodology VMR0006, ver. 1.1. The values are taken as weighted averages of the crediting period vintage for the number of days in the monitoring period. The calculation is explained in the ER Calculation Sheet. Here, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.		

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove
Value applied:	0.1023
Comments	Calculated using the Default value as specified in VMR0006 methodology and the data obtained through the distribution surveys.

Data / Parameter	μ_y
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Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Value applied:	PAI-1: 0.8172
Comments	N/A

6.2 Baseline Emissions

Year	Baseline emissions or removals (tCO ₂ e)
01/01/2021 – 31/12/2021	456,834
01/01/2022 – 30/06/2022	226,539
Total	683,373

6.3 Project Emissions

Not applicable, since the applied methodology does not have provisions for separate calculation of project emissions.

6.4 Leakage

Separate representation of leakage is not applicable in this case, since the applied methodology does not have provisions for separate calculation of leakage emissions. The leakage is incorporated as default within the ER calculations, in-line with the application of net to gross adjustment factor to account for the leakage, with the value 0.95. This is evident from the applied methodology VMR0006, ver. 1.1, Section 8.3.

6.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01/01/2021 – 31/12/2021	456,834	-	-	456,834

01/01/2022 – 30/06/2022	226,539	-	-	226,539
Total	683,373	-	-	683,373